PULSE: Payment Understanding, Lifecycle Scoring & Embeddings
A Transaction Foundation Model on Synthetic ISO 20022 Rails - a Falsifiable Replication and Extension of arXiv:2410.07851

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Preprint · July 2026

Abstract. We present PULSE, a transaction foundation model built by replicating the architecture of Raman et al. (arXiv:2410.07851) on synthetic ISO 20022 payments and measuring, against thresholds fixed before each run, what a single frozen tabular encoder can and cannot predict about a payment. Five claims were pre-registered. Two replicate the source paper: the partitioned high-cardinality embedder matches dense masked-column reconstruction at 5.8% of the embedding parameters (C1), and the frozen-encoder-plus-adapter stack trains 0.59% of a full fine-tune's parameters but loses to CatBoost by 44.5 PR-AUC points on a rule-generated risk label (C2) - trees read rule labels directly; the encoder bottleneck cannot. Three claims concern sequences. A small history encoder over frozen per-row embeddings beats an order-blind pooled baseline by 38.5 PR-AUC points (C3) and CatBoost on hand-built aggregates by 41.3 points (C4) on a regime-change task whose order-invariant statistics are matched across classes. Feeding the same history representation through a frozen Phi-1.5 with 7.6M trainable adapter parameters buys 1.3 points over a linear probe (C5) - so the LLM is not required for fixed-menu accuracy, though it remains the serving interface for reasons C5 does not measure. We then extend the corpus to a nine-message ISO 20022 lifecycle (pain.001 through camt.056/pacs.004) and report, wins and nulls alike: settlement-time regression beats its naive baseline at initiation (MAE 172 vs 234 minutes), in-flight outcome scoring sharpens from 0.86 to 0.94 PR-AUC as messages arrive, and fusing pain.001 origination context lifts account-takeover detection from chance (0.03) to 1.00; message-order signal is a measured null (-0.5 points), and cancel/return heads sit at prevalence. Two follow-up interventions on the same frozen encoder - LoRA adapters (closing 40.7% of the tree gap) and full-token read-outs (+72%, still below the adapters) - independently confirm that complete-row rule labels belong to the tree (C9, C10). All data is synthetic and seeded; 209 tests reproduce every number.

1 Introduction

A payments engine asks many questions about the same row. Which rail should carry this transfer? Will it settle straight through, and if not, which exception fires? When does the beneficiary see funds? Is this account suddenly bursting with activity it never showed before? Production systems typically answer each question with its own model, its own features, and its own retraining calendar. Raman et al. [1] propose the alternative this paper tests: pretrain one tabular encoder on the payment corpus, freeze it, and answer every downstream question with a small adapter or probe on the frozen representation.

Replications in this space tend to fail quietly - a retuned learning rate here, a relabeled dataset there, and the headline number survives while the claim it supported dies. We took the opposite discipline. Every architectural choice is pinned to the source paper's section that mandates it, thresholds live in a config file committed before the runs, and a miss stays a miss: no post-hoc retuning closed any gap reported here. The result is a ledger rather than a victory lap. Some claims replicate strongly. One fails against our own bar, and the failure is informative. One resolves a design fork - whether the frozen LLM earns its inference cost - with a number instead of an opinion.

Our contributions: (i) an independent replication of the source paper's parameter-efficiency claims at near-paper vocabulary scale (119,819 accounts) on fully synthetic data; (ii) three pre-registered sequence claims measured on a corpus whose order-invariant statistics are matched across classes, isolating temporal signal by construction; (iii) a measured answer to the frozen-LLM question - 1.3 PR-AUC points for 7.6M adapter parameters plus a 1.3B-parameter forward pass; (iv) an extension of the single-table setting to a nine-message ISO 20022 lifecycle with event-anchored timestamps, recall and return legs, and a second data source (pain.001 origination context) whose fusion we evaluate; and (v) a capability-routing discipline that names which questions the model refuses to answer because a rule, a lookup, or a template answers them exactly.

2 Related work

Transaction representation learning has moved from sequence models over card streams - TabBERT [7], CoLES [8], FATA-Trans [9] - toward foundation-model framings in which one pretrained backbone serves many tasks: TransactionGPT [10] at Visa and PRAGMA [11] at Revolut are the production-scale statements of that position. Raman et al. [1], our source, contribute the tabular side: columns as tokens, a partitioned embedder for high-cardinality identifiers, an offline party encoder, and a frozen LLM decoder adapted with small trainable modules in the prefix-tuning family [4]. The C2 finding we report - gradient-boosted trees [6] winning on labels that are rules over raw features - repeats a decade of tabular-benchmark results and is why we treat CatBoost as the bar every claim must clear, not as a strawman. TabPFN [12] offers a different foundation-model bet (in-context learning over small tables); we assessed it as out of scope because our tables exceed its row and class budgets and our deployment needs a persistent, incrementally probed representation.

3 Synthetic corpora

No real transaction data appears anywhere in this work. Three generators, all seeded and committed, produce the corpora; identical commands reproduce byte-identical files on any machine.

3.1 pacs.008 corpus (claims C1, C2)

One million interbank credit transfers over 19,895 corporate parents and a combined account vocabulary of 119,819 - close to the source paper's ~125K. The risk label is deliberately a transparent rule over raw fields (cross-border flag, currency, region, industry, amount bands), with the High class at 1.8% prevalence. That choice is honest hostility: a label a tree can read directly is the hardest case for an encoder that must compress the row into one vector before any head sees it.

3.2 Behavioural corpus (claims C3-C5)

Per-account payment histories, 8 to 40 events each, where 45% of accounts undergo a regime change midway through their stream. The generator matches the gap multiset and the amount distribution across the two classes, so every order-invariant aggregate - means, quantiles, counts, totals - carries no class information. Only a model that reads the ordered, timed sequence can separate the classes. This construction is what lets us attribute C3 and C4 to temporal signal rather than to feature leakage. The full corpus yields 24,016 sequences; evaluation holds out 20% of accounts (4,803 sequences, positive prevalence 0.450), never rows within a training account.

3.3 India-rails lifecycle corpus (Section 6)

Twenty thousand payments across RTGS, NEFT, IMPS, and SWIFT, each expanded into its ISO 20022 message trail: pain.001 acceptance, pain.002 status, pacs.008 clearing (pacs.009 cover where correspondent banking requires it), pacs.002 as the gpi tracker backbone, camt.054 credit notification, camt.056/camt.029 recall legs, and pacs.004 returns. Message timestamps anchor to generated exception events rather than to interpolation, and each message carries a direction and a visibility flag - an inward payment's pain.* legs live at the remote bank and never enter our prefixes, which is the information set a real engine has. A separate origination table attaches channel, device, session, and authentication attributes to each outward pain.001 at t=0 (10,943 rows), simulating the data a bank's own channel layer possesses before any ISO message exists. A cadence-enabled variant of the same generator (salary, rent, utility, supplier schedules) supports the forecasting heads.

3.4 How the generators work

All three corpora come from the same recipe, and the recipe is the falsifiability tool: because every label has a known cause, a claim can fail loudly. Generation runs in five steps. First, an account universe: 4,000 synthetic corporates with industries, countries, and account/IFSC/BIC identifiers (20,000 parents at the 1M-row scale).

Second, payments: amount and scope select the eligible rail - 2 lakh INR unlocks RTGS, amounts at or under 5 lakh allow IMPS, cross-border routes to SWIFT - and a deliberate slice arrives mis-routed so the routing head has something real to correct. Third, exceptions: a hazard model injects sixteen exception types (sanctions holds, liquidity shortfalls, format errors, and so on) at specific lifecycle steps, and terminal status, settlement time, and charges follow from the exception path and the rail's service curve. Fourth, lifecycle expansion: each payment becomes its ISO 20022 message trail, with timestamps anchored to the injected events rather than interpolated - a return leg lands 60 minutes after its cause, a recall 240. Fifth, planted couplings: 3% of payments receive cancellation requests, origination-context account-takeover risk is coupled (roughly eightfold) to later recalls, and an optional cadence mixture (salary, rent, utility monthly; supplier weekly) gives the forecasting heads a periodicity to find. Every generator is seeded: the same command produces byte-identical corpora on any machine, which is what lets a single results file stand as the record.

3.5 What a deployment must supply

The serving contract is deliberately narrow. Intake scoring needs nineteen fields a payments engine already holds at acceptance: debtor and creditor accounts, ultimate parties, amount and currency, settlement date and method, identifier type, party names, countries, and industry codes. In-flight scoring needs the message trail so far - end-to-end id, sequence number, message type, transaction status, and minute offsets - which is the engine's own event log. The behavioural endpoints (velocity, next-payment) are stateless: the engine sends each account's recent history (two or more rows for velocity, three or more for forecasting) and PULSE holds nothing between calls, which keeps the model out of the data-retention conversation. The origination fusion path additionally reads the bank's own channel-layer attributes at pain.001 time - channel, device, session, and authentication signals that exist before any ISO message does. Nothing in the contract requires enrichment beyond what the engine and channel layer already produce.

4 Architecture

The pinned encoder follows the source paper: each column embeds as one token (25M parameters total), high-cardinality identifiers route through the partitioned embedder of their Section 3.1, party identities come from an offline-pretrained persistent store, and amounts pass through a currency-conditioned adaptive quantizer. Pretraining combines masked-column reconstruction with a batch-hard triplet objective [2], three epochs, then the encoder freezes for good. Nothing downstream ever updates it.

For sequences, a small transformer (the history encoder) consumes the frozen per-row embeddings of an account's ordered payments together with time deltas and calendar features, pretrained by reconstructing held-out field values, and emits one history vector per prefix. Downstream heads then take one of two forms. Option A is a linear probe on the history vector. Option B routes the same vector into a frozen Phi-1.5 [5] through trainable adapters in the prefix family [3, 4] - 7,638,528 trainable parameters against the LLM's 1.3B frozen ones (Figure 1). C5 is the measured comparison between the two.

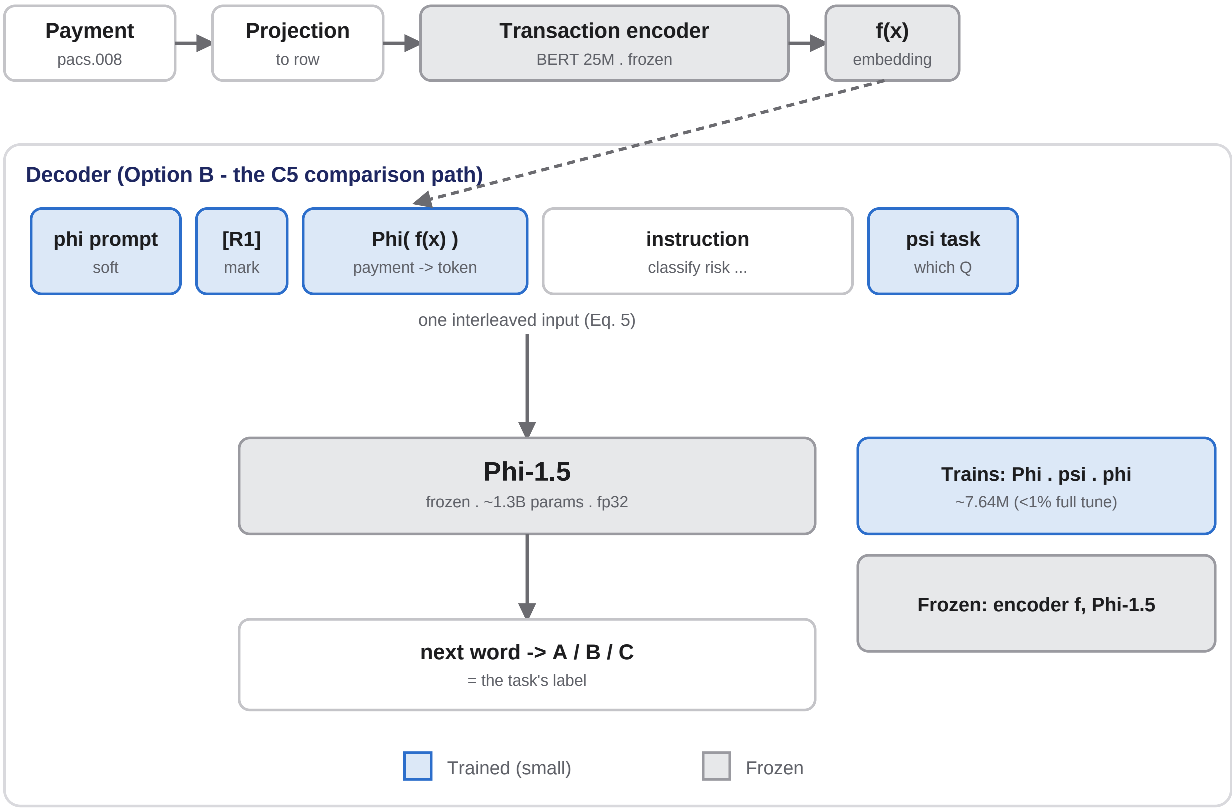

Figure 1: the frozen-encoder, frozen-LLM decoder with pretrained adapters, redrawn from the architecture of Raman et al. [1] (Section 4, Eq. 5); our own rendering, not a reproduction of their figure. Grey boxes never train; blue boxes are the trainable trio - the adapter Phi projecting f(x) into the LLM's embedding space, the task vector psi, and the soft prompt phi. The frozen LLM reads the interleaved sequence as inputs_embeds and its next-word distribution, restricted to the answer tokens, is the prediction. This decoder is PULSE's serving interface for the classification menu (Section 7); C5 measures what it costs against a probe.

Not every question reaches the model. Rail caps and floors, assigned reference values, identifier lookups, and duplicate checks are answered by guards and templates that are exact by construction; we delisted those tasks from the model's advertised capabilities after finding early prototypes claiming credit for predictions a two-line rule makes with certainty. The serving layer routes each field to model, rule, lookup, or template, and the documentation names which is which. A model that answers everything answers some things worse than the infrastructure it runs on.

5 Pre-registered claims

Table 1 states each claim, its threshold as committed in configuration before the run, and the measured outcome. Full-scale numbers come from single seeded runs on one NVIDIA H200; smoke-scale runs on CPU reproduce every qualitative verdict.

Claim	Measured	Threshold	Verdict
C1 partitioned embedder: parameter ratio vs dense, at matched reconstruction	**0.058**; recon gap -0.012 pp (partitioned marginally better)	ratio <= 0.55; gap <= 1 pp	**pass**
C2a adapter parameter efficiency	**0.0059** of full fine-tune (7.64M / 1.3B)	<= 0.10	**pass**
C2b adapter beats CatBoost on rule-label risk	PR-AUC 0.210 vs **0.655** (-44.5 pp)	>= +10 pp	**fail**
C3 temporal lift: history encoder vs order-blind pooling	0.919 vs 0.533 (**+38.5 pp**)	>= +10 pp	**pass**

Claim	Measured	Threshold	Verdict
C4 held-out generalisation vs CatBoost on aggregates	0.919 vs 0.506 (**+41.3 pp**)	>= +5 pp	**pass**
C5 LLM necessity: frozen Phi-1.5 + adapters vs linear probe	0.932 vs 0.919 (**+1.3 pp** for the LLM)	drop the LLM if within 2 pp	**drop the LLM**
C9 adapter capacity: LoRA inside the frozen encoder vs probe vs CatBoost	probe 0.152, LoRA **0.301**, CatBoost 0.517	close >= 50% of the probe-to-tree gap	**fail** (40.7%)
C10 read-out bandwidth: all-token concat vs the single row token, no adapters	single 0.152, mean 0.163, concat **0.261** - below the LoRA reference	close >= 50% of the LoRA-left gap	**fail**

Table 1: the claims. C1/C2 on the 1M-row pacs.008 corpus (positive class 1.8%); C3-C5 on the behavioural corpus, held-out accounts, PR-AUC on regime change; C9/C10 a matched pair on a 20k CPU slice of the pacs.008 corpus (Section 5.4). Thresholds predate the runs; no configuration was retuned after a miss.

5.1 *What replicated*

C1 is the source paper's core engineering claim and it replicates cleanly: the partitioned embedder holds reconstruction quality while spending 5.8% of the dense embedding budget, at a vocabulary within a few percent of the paper's. C3 and C4 are the strongest results in this report. On data where every order-invariant aggregate is uninformative by construction, CatBoost sits near chance (0.506) while the history encoder reaches 0.919 on unseen accounts - the first configuration in this project where the learned representation beats the tree, and it does so because the signal is genuinely temporal. A companion velocity task (does this account's next window contain an activity burst, prevalence 0.131) shows the same shape: time-aware 0.822 against order-blind 0.222.

5.2 *What failed, and why we kept it*

C2b failed against our own bar. The adapter learns real signal - 0.210 PR-AUC is roughly ten times the prevalence floor - but CatBoost reads the label-generating rule straight off the raw features, and no amount of frozen-representation elegance competes with direct access to the rule. We could have made this number respectable by softening the label. We left it, because it draws the deployment boundary honestly: on complete rows with rule-like labels, ship the tree. The encoder earns its keep where trees cannot follow - order, timing, partial information, and fusion across sources - which is precisely where Sections 5.1 and 6 place it.

5.3 *The LLM verdict - and its limits*

C5 asks a narrow question: does the frozen LLM add accuracy on one fixed task? On regime classification from the same frozen history vector, Option B (Phi-1.5 plus adapters) reaches 0.932 against the probe's 0.919 - a real but small gap of 1.3 points, under the 2-point bar we set in advance, purchased with 7.6M trained parameters and a 1.3B-parameter forward pass per prediction. An earlier mock-LLM control had suggested the LLM subtracts 13 points; the real model corrects the sign but not the decision. We state the scope plainly, because we initially overread this result ourselves: C5 is our question, not the source paper's. Their claim for the decoder is the interface - one frozen LLM answers every task through instructions, a new question costs an instruction string plus a tiny task vector, records interleave for multi-record reasoning, and the output space is language. A linear probe tests none of that. C5 therefore licenses exactly one conclusion: the LLM is not required for fixed-menu accuracy. It does not license removing the decoder, and Section 7 describes the serving architecture that keeps it.

5.4 *What the freeze costs (C9), what bandwidth buys (C10)*

C2b left an open question: was the 44.5-point loss to CatBoost the price of the frozen encoder, of the adapter family, or of the read-out contract - the whole row compressed into one 512-d vector before any head sees it? Two follow-up interventions on the SAME frozen encoder, run as a matched pair on a 20k-row CPU slice, decompose it. C9 enriches the adapter family: low-rank (LoRA) adapters inside the encoder's feed-forward layers - the base weights frozen structurally, zero-initialised so step 0 is exactly the pretrained model, 328k trainable parameters. Converged with early stopping, LoRA doubles the linear probe (0.152 to 0.301) yet closes only 40.7% of the gap to CatBoost (0.517), under the pre-registered 50% bar. C10 widens only the read-out: a linear head over all eleven token outputs instead of the row token, no adapters, 5.6k head parameters. Full-bandwidth concat reaches 0.261 - a 72% relative gain for zero backbone training, and still below the converged LoRA arm.

The pair says something one intervention alone could not: adapter capacity and read-out bandwidth each recover real signal, and neither - nor plausibly their sum - reaches a tree that reads the label-generating rule off the raw features. C2's deployment verdict is thereby confirmed by two independent interventions rather than asserted once: complete-row rule labels belong to the tree, and the twin's capability routing sends them there. The learned representation's wins stay exactly where Sections 5.1 and 6 measured them - order, timing, partial information, and fusion. Caveats attach: the pair ran at 20k rows (C2's canonical scale is 1M), the encoder was pretrained on the same slice, and torch's fused attention restricts LoRA to the feed-forward linears.

6 The lifecycle extension: wins and nulls

The India-rails corpus turns the single-table problem into an engine's problem: score at intake, rescore as each message lands, and answer questions no single message contains. Table 2 reports every prediction the system exposes, including the ones that lose. Splits hold out accounts; in-flight prefixes exclude the messages that state the outcome being predicted.

Prediction	Measured	Baseline	Reading
Settlement ETA at initiation (pain.001 only)	MAE **172 min**	234 (naive)	win, pre-clearing
Settlement ETA at intake (full row, n=4,000 eval)	MAE **195 min**	392 (naive)	win
Charges	MAE **117**	203 (naive)	win
'Will it credit?' as messages arrive	PR-AUC 0.860 -> **0.942** (prefix 1 -> 3)	0.856 (prevalence)	win, sharpens in-flight
Account takeover at origination	0.029 ISO-only -> **1.000** fused with pain.001 channel context	ISO fields alone	win, multi-source
Velocity burst (dense fleet)	PR-AUC **0.822**	0.222 order-blind	win, timing-only
Next payment: days until (held-out customers)	MAE **23.5 d**	25.9 / 32.5 (median / last gap)	win, modest
Next payment: occurrence within 35 d	PR-AUC 0.579	tree 0.655, rule 0.637	loss - trees win
Next payment: amount	MAE 33.0k	21.8k (history median)	loss
Rail routing	acc 0.556	tree 0.645, majority 0.478	loss - C2 territory (SWIFT F1 = 1.00)
Terminal status at intake	acc 0.279	majority 0.740	loss - C2 territory
Reject reason (ISO code)	acc 0.102	'none' dominates	loss
Cancel / return likelihood at intake	PR-AUC at prevalence (0.030 / 0.018)	-	null - event-driven, not intent-visible
Message-order signal (tracker outcome)	sequence macro-F1 0.270 vs pooled 0.275 (**-0.5 pp**)	bag of messages	null - prefixes are 2-3 near-fixed-order messages

Table 2: every exposed prediction on the lifecycle corpus, H200 full runs. The losses and nulls are load-bearing: they route those fields to trees, rules, or nothing.

Two rows deserve prose. The account-takeover result is the multi-source argument in one line: ISO clearing fields carry no trace of a hijacked session (0.029, chance at 2.7% prevalence), while the same frozen encoder reading the pain.001 origination context alongside them separates the classes completely - on synthetic data whose generator plants exactly that correlation, so the number demonstrates the fusion path works, not that real fraud is this easy. The message-order null is the opposite lesson. A single payment's pre-outcome trail is two or three messages in near-fixed order; there is nothing for a sequence model to read that a bag of messages does not already say, and the measured -0.5 points says exactly that. Temporal signal in payments lives at the entity level (C3, velocity), not inside one payment's message chain.

7 Serving

An earlier revision of this system served every question from its own linear head - roughly ten per-task probes plus calibrators on the frozen encoder. That was a quiet departure from the source architecture: the paper's decoder makes a new question cost an instruction string, while a head farm makes it cost a new head, a refit, and a recalibration. We corrected it. The serving interface for the classification menu (rail, status, risk, geography, expense, and settlement time as a banded class) is now the paper's decoder itself: one frozen LLM plus the trained adapter trio, instruction-conditioned exactly as in Figure 1, answering with a softmax over answer tokens. The per-task classification probes and their calibrators are not shipped in this configuration.

Two families of question stay on probes, as documented boundaries rather than hidden ones. Regression - the ETA point estimate that feeds the liquidity curve - has no mechanism in an answer-token decoder beyond banding, so the banded class rides the decoder while the point estimate stays a Ridge head. The rare-event probability-of-default heads (cancel, return, the sixteen exception codes) and the sequence extensions (in-flight, velocity, forecasting) are our additions, outside the paper's single-record scope, and keep their calibrated probes. Scoring remains stateless throughout: the engine sends rows, prefixes, or histories and holds all state. Two operational caveats are measured rather than suspected: brand-new accounts degrade routing confidence to near-uniform (identity features carry most of the rail signal), and a bundle trained under scikit-learn 1.9 needs a one-line attribute shim to load under 1.7. A pre-registered follow-up (C6 in the project ledger) measures what fidelity costs: the decoder's per-task answer quality against the retired probes on the same held-out split, at full scale with the real Phi-1.5.

8 Limitations

Everything here is synthetic, and the generators were written by the same team that wrote the models. We defended against self-deception where we knew how - matched aggregates for C3/C4, prevalence-honest PR-AUC, held-out actors, visibility-filtered prefixes, thresholds committed in advance - but a generator cannot invent the correlations of real payment traffic, and the ATO fusion number in particular measures a planted signal. Each full-scale number is one seeded run; we report no variance across seeds. C5 is measured on one task family with one 1.3B LLM, and says nothing about text-emitting or few-shot settings. The rare-event heads (cancel at 3.0%, return at 1.8% of 20,000 payments) sit at prevalence and would need two orders of magnitude more data before failure means anything. The published v1 checkpoint predates the lifecycle work and scores only the original task suite. Real transaction data, multiple seeds, and an entity-level generator whose successive payments correlate are the three shortest paths to making these numbers mean more.

9 Conclusion

PULSE's frozen encoder does carry many payment decisions - provided one is honest about which ones. Where the signal is a rule over a complete row, a gradient-boosted tree wins on accuracy (C2b, routing, status) and we report it. Where the signal is order, timing, partial information, or a second source the tree never sees, the frozen representation wins by margins of 38 to 60 PR-AUC points (C3, C4, velocity, in-flight, fusion). The frozen LLM contributes 1.3 accuracy points on a fixed task (C5) - and stays, because the source paper's argument for it was never accuracy on a fixed task: it is the interface that prices a new question at one instruction string, and PULSE serves it as such. Where the label is a rule over a complete row, two further interventions - richer adapters (C9) and wider read-outs (C10) - each recover real signal and still lose to the tree, so the routing verdict is confirmed, not assumed. The claims ledger, generators, and 209 tests reproduce every number in this paper from two commands.

Annexure A Live API round-trips

Every block below is a real request/response pair captured against the served bundle (api_india:app, GPU-trained heads, isotonic-calibrated). The scenarios were picked to be distinct, and two of them are deliberate stress cases: a mis-routed payment and a pair of accounts the model has never seen. docs/annex_capture.json holds the unabridged captures; the capture script reruns them against any bundle.

A.1 Intake: POST /score/intake - five scenarios, one endpoint

The first request is shown in full; the rest send the same 19 fields with different values. Ground truth from the held-out generator run appears with each response - misses included.

UC-1, small domestic transfer (833 INR, known accounts). True rail IMPS, true status STP:


```
{
 "request": {
  "DbtrAcct_Id": "AZLN7144",
  "CdtrAcct_Id": "0QF6OXNV",
  "UltmtDbtr_Id": "HT8JC5",
  "UltmtCdtr_Id": "UWK0LS",
  "IntrBkSttlmAmt": 832.91,
  "Ccy": "INR",
  "IntrBkSttlmDt": "2023-11-30",
  "SttlmMtd": "CLRG",
  "identifier_type": "ACCT_IFSC",
  "Dbtr_Nm": "Granite GmbH Capital Services",
  "Cdtr_Nm": "Harbor Inc Sales Capital",
  "UltmtDbtr_Nm": "Granite GmbH Capital",
  "UltmtCdtr_Nm": "Harbor Inc Sales",
  "Dbtr_Ctry": "IN",
  "Cdtr_Ctry": "IN",
  "Dbtr_Industry": "Communications",
  "Cdtr_Industry": "Energy",
  "Dbtr_SubIndustry": "Media",
  "Cdtr_SubIndustry": "Renewables"
 },
 "response": {
  "rail_pred": "IMPS",
  "rail_conf": 0.624,
  "status_pred": "STP",
  "eta_min_pred": 0.0,
  "risk_pred": "Low",
  "geography_pred": "Asia",
  "expense_pred": "Operational",
  "top_exception_risks": [
   {
    "code": "format_error",
    "score": 0.04
   },
   {
    "code": "below_min",
    "score": 0.04
   },
   {
    "code": "fraud_hold",
    "score": 0.03
   }
  ]
 }
}
```


UC-2, high-value domestic (250,799 INR). Rail and risk correct (RTGS 0.812, High); the status head calls REJECTED where truth is STP - the weak head of Table 2, shown failing:


```
{
 "rail_pred": "RTGS",
 "rail_conf": 0.812,
 "status_pred": "REJECTED",
 "eta_min_pred": 28.3,
 "risk_pred": "High",
 "geography_pred": "Asia",
 "expense_pred": "Capital",
 "top_exception_risks": [
  {
   "code": "fraud_hold",
   "score": 0.04
  },
  {
   "code": "format_error",
   "score": 0.04
  },
  {
```

```
   "code": "below_min",
   "score": 0.04
  }
 ]
}
```


UC-3, cross-border (BIC/IBAN, COVE settlement, IN -> BR). SWIFT at confidence 1.00; ETA 878 min against a true settle time of 1,140:


```
{
 "rail_pred": "SWIFT",
 "rail_conf": 1.0,
 "status_pred": "REJECTED",
 "eta_min_pred": 878.2,
 "risk_pred": "High",
 "geography_pred": "Asia",
 "expense_pred": "Capital",
 "top_exception_risks": [
  {
   "code": "sanctions_hit",
   "score": 0.05
  },
  {
   "code": "format_error",
   "score": 0.04
  },
  {
   "code": "settlement_fail",
   "score": 0.03
  }
 ]
}
```


UC-4, mis-routed below the RTGS floor (7,410 INR arrives tagged RTGS; the 2-lakh floor makes it invalid). The model re-routes to NEFT and predicts the reject - both correct:


```
{
 "rail_pred": "NEFT",
 "rail_conf": 0.631,
 "status_pred": "REJECTED",
 "eta_min_pred": 235.6,
 "risk_pred": "Low",
 "geography_pred": "Asia",
 "expense_pred": "Operational",
 "top_exception_risks": [
  {
   "code": "below_min",
   "score": 0.06
  },
  {
   "code": "sanctions_hit",
   "score": 0.04
  },
  {
   "code": "format_error",
   "score": 0.04
  }
 ]
}
```


UC-5, cold start (every account id brand new). The confidence collapse is the documented behaviour, and the reason routing confidence is part of the response contract - a gateway should treat 0.001 as 'do not trust this rail call':


```
{
 "rail_pred": "NEFT",
 "rail_conf": 0.001,
 "status_pred": "REPAIRED",
 "eta_min_pred": 1360.6,
 "risk_pred": "Medium",
 "geography_pred": "Asia",
 "expense_pred": "Other",
 "top_exception_risks": [
  {
   "code": "sanctions_hit",
   "score": 0.1
  },
  {
   "code": "missing_field",
   "score": 0.07
  },
  {
   "code": "fx_fail",
```

```
   "score": 0.05
  }
 ]
}
```

A.2 Explanations: explain:true - column-occlusion drivers

Drivers for UC-3. identifier_type (BIC_IBAN) carries most of the risk call; currency and settlement method carry the rail call. Occlusion is faithful to the frozen encoder by construction - no post-hoc surrogate model:

```
[
 {
  "field": "identifier_type",
  "rail_impact": 0.2905,
  "risk_impact": 0.5831,
  "eta_impact_min": 197.1
 },
 {
  "field": "Ccy",
  "rail_impact": 0.4739,
  "risk_impact": -0.2017,
  "eta_impact_min": 119.9
 },
 {
  "field": "IntrBkSttlmAmt",
  "rail_impact": -0.0059,
  "risk_impact": 0.5572,
  "eta_impact_min": -276.9
 },
 {
  "field": "IntrBkSttlmDt",
  "rail_impact": -0.0048,
  "risk_impact": -0.2776,
  "eta_impact_min": -114.6
 },
 {
  "field": "UltmtCdtr_Id",
  "rail_impact": -0.0034,
  "risk_impact": 0.1949,
  "eta_impact_min": -107.8
 }
]
```

A.3 In-flight: POST /score/inflight - healthy vs troubled lifecycle

UC-7 scores a payment from its first visible message only. UC-8 scores a lifecycle that went wrong end to end: settled (pacs.002 ACSC), booked (camt.054), then a recall (camt.056), the recall resolution (camt.029 CNCL) and the return of funds (pacs.004). Return probability rises tenfold over the healthy case and the conditional reject reason snaps to AC04 (account closed) at 0.97:

```
{
 "UC7_prefix": [
  "pacs.008(None)"
 ],
 "UC7_response": {
  "end_to_end_id": "E2E-00020000",
  "n_msgs": 1,
  "last_msg_type": "pacs.008",
  "booked_proba": 0.9571,
  "settlement_outcome": {
   "MANUAL_REVIEW": 0.0739,
   "REJECTED": 0.0248,
   "REPAIRED": 0.1538,
   "STP": 0.7475
  },
  "eta_remaining_min": 1.0,
  "reject_reason_if_failed": {
   "code": "ED05",
   "score": 0.5882
  },
  "cancel_proba": 0.0259,
  "return_proba": 0.0063
 }
}
```

```
{
 "UC8_prefix": [
  "pacs.008(None)",
  "pacs.002(ACSC)",
  "camt.054(BOOK)",
  "camt.056(None)",
  "camt.029(CNCL)",
  "pacs.004(None)"
 ],
```

```
 "UC8_response": {
  "end_to_end_id": "E2E-00020005",
  "n_msgs": 6,
  "last_msg_type": "pacs.004",
  "booked_proba": 0.8522,
  "settlement_outcome": {
   "MANUAL_REVIEW": 0.0057,
   "REJECTED": 0.0575,
   "REPAIRED": 0.1289,
   "STP": 0.808
  },
  "eta_remaining_min": 1.5,
  "reject_reason_if_failed": {
   "code": "AC04",
   "score": 0.9665
  },
  "cancel_proba": 0.044,
  "return_proba": 0.0625
 }
}
```


A.4 Behaviour and treasury: velocity, next-payment, liquidity

UC-9 sends two engine-supplied account histories to POST /score/velocity; the burst score separates them while the transparent gap rule (returned alongside for comparison) fires on neither. UC-10 sends a salaried customer and an irregular one to POST /forecast/next - the cadence account gets the higher occurrence probability and the shorter expected gap. UC-11 sends the day's 100-payment book to POST /forecast/liquidity and receives the outflow curve by rail and settlement-time bucket; bucket sums equal the book total by construction:


```
{
 "UC9_velocity": {
  "bursting": {
   "actor": "1RR9184S",
   "n_txns": 6,
   "burst_proba": 0.5,
   "burst_rule": 0
  },
  "quiet": {
   "actor": "001IR497",
   "n_txns": 6,
   "burst_proba": 0.0,
   "burst_rule": 0
  }
 }
}
```

```
{
 "UC10_forecast_next": [
  {
   "actor": "F1BBU9MT",
   "n_history": 20,
   "next_event_proba": 0.75,
   "expected_gap_days": 9.9,
   "expected_amount": 49204.2617,
   "top_payee_hint": "NU5WVQSZ",
   "horizon_days": 35
  },
  {
   "actor": "KNAGDL0J",
   "n_history": 12,
   "next_event_proba": 0.6103,
   "expected_gap_days": 26.1,
   "expected_amount": 13660.1797,
   "top_payee_hint": "X09NS2VJ",
   "horizon_days": 35
  }
 ]
}
```

```
{
 "UC11_liquidity": {
  "buckets": [
   "<1m",
   "1-15m",
   "15-60m",
   "60-240m",
   "240-1440m",
   ">1440m"
  ],
  "by_rail": {
   "IMPS": [
    37050.91,
    0.0,
```

```
      3975.88,
      14208.57,
      176610.42,
      51850.99
     ],
     "NEFT": [
      113832.09,
      178928.58,
      0.0,
      54790.55,
      602484.36,
      62716.79
     ],
     "RTGS": [
      0.0,
      0.0,
      250799.11,
      0.0,
      436447.05,
      200199.28
     ],
     "SWIFT": [
      0.0,
      0.0,
      0.0,
      0.0,
      1093808.58,
      380952.58
     ]
    },
    "total": [
     150883.0,
     178928.58,
     254774.99,
     68999.12,
     2309350.41,
     695719.64
    ],
    "n_payments": 100,
    "total_amount": 3658655.74
   }
  }
```


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